EXPERIMENT NO -3

NAME:Dhanush v Roll no-240701115

1. To-Do List Program using CLI

```
# To-Do List Program (CLI)

tasks = []

def add_task(task):
    tasks.append(task)
    print(f"Task '{task}' added.")

def view_tasks():
    if tasks:
        print("Your tasks:")
        for idx, task in enumerate(tasks, 1):
            print(f"{idx}. {task}")
    else:
        print("No tasks to show.")

def remove_task(task_number):
    if 0 < task_number <= len(tasks):
        removed_task = tasks.pop(task_number - 1)
        print(f"Task '{removed_task}' removed.")
    else:
        print("Invalid task number.")

def main():
    while True:
        print("\nOptions:")
        print("1. Add Task")
        print("2. View Tasks")
        print("3. Remove Task")
        print("4. Exit")

        choice = input("Enter your choice: ")

        if choice == '1':
            task = input("Enter task: ")
            add_task(task)
        elif choice == '2':
            view_tasks()
        elif choice == '3':
            task_number = int(input("Enter task number to remove: "))
            remove_task(task_number)
        elif choice == '4':
            print("Exiting...")
            break
        else:
            print("Invalid choice. Please try again.")

main()
```

2. To-Do List Program using Gui (Second Version)

```
# To-Do List Program (CLI - Duplicate Version)

tasks = []
def add_task(task):
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    tasks.append(task)
    print(f"Task '{task}' added.")

def view_tasks():
    if tasks:
        print("Your tasks:")
        for idx, task in enumerate(tasks, 1):
            print(f"{idx}. {task}")
    else:
        print("No tasks to show.")

def remove_task(task_number):
    if 0 < task_number <= len(tasks):
        removed_task = tasks.pop(task_number - 1)
        print(f"Task '{removed_task}' removed.")
    else:
        print("Invalid task number.")

def main():
    while True:
        print("\nOptions:")
        print("1. Add Task")
        print("2. View Tasks")
        print("3. Remove Task")
        print("4. Exit")

        choice = input("Enter your choice: ")

        if choice == '1':
            task = input("Enter task: ")
            add_task(task)
        elif choice == '2':
            view_tasks()
        elif choice == '3':
            task_number = int(input("Enter task number to remove: "))
            remove_task(task_number)
        elif choice == '4':
            print("Exiting...")
            break
        else:
            print("Invalid choice. Please try again.")

main()
```

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3. To-Do List Program using Speech Recognition

```
# To-Do List Program using Speech Recognition

import speech_recognition as sr
import pyttsx3

tasks = []

recognizer = sr.Recognizer()
engine = pyttsx3.init()

def add_task(task):
    tasks.append(task)
    engine.say(f"Task {task} added")
```

```
    engine.runAndWait()

def view_tasks():
    if tasks:
        engine.say("Your tasks are")
        for i, task in enumerate(tasks, 1):
            engine.say(f"Task {i} is {task}")
    else:
        engine.say("No tasks to show")
    engine.runAndWait()

def remove_task(task_number):
    if 0 < task_number <= len(tasks):
        removed_task = tasks.pop(task_number - 1)
        engine.say(f"Task {removed_task} removed")
    else:
        engine.say("Invalid task number")
    engine.runAndWait()

def recognize_speech():
    with sr.Microphone() as source:
        print("Listening...")
        audio = recognizer.listen(source)

    try:
        command = recognizer.recognize_google(audio)
        print("You said:", command)
        return command.lower()
    except sr.UnknownValueError:
        engine.say("Sorry, I did not understand")
        engine.runAndWait()
        return None

def main():
    while True:
        engine.say("Say add task, view tasks, remove task, or exit")
        engine.runAndWait()

        command = recognize_speech()

        if not command:
            continue

        if "add task" in command:
            engine.say("What is the task?")
            engine.runAndWait()
            task = recognize_speech()
            if task:
                add_task(task)

        elif "view tasks" in command:
            view_tasks()

        elif "remove task" in command:
            engine.say("Say task number")
            engine.runAndWait()
            number = recognize_speech()
            if number and number.isdigit():
                remove_task(int(number))

        elif "exit" in command:
            engine.say("Exiting")
            engine.runAndWait()
            break

        else:
            engine.say("Invalid option")
            engine.runAndWait()
```

```
if __name__ == "__main__":
    main()
```